

Pose investigable questions to identify patterns and test relationships and make reasoned predictions

Learning intention

- pose investigable questions to identify patterns and test relationships and make reasoned predictions.

Teach and model

- discussing and refining questions to enable scientific investigation.
- making reasoned predictions about electrical circuit function based on a picture or diagram of a circuit.
- posing investigable questions to test relationships, such as: 'Will more salt dissolve in warm water than in cold water?'.

Check understanding

- Explain the main idea and give one accurate example.
- Show the idea in a second way: with words, a model, a diagram or symbols.
- Apply the idea to a short unfamiliar situation and justify the result.

Teaching cautions

- Ask the student to explain their choice, not only state an answer.
- Check that key vocabulary is understood in a new example.
- Mix representations so the skill is transferred rather than copied.

Quick lesson sequence

- Connect to prior knowledge and introduce the key vocabulary.
- Model one clear example, then solve a second example together.
- Finish with an independent check and a short student explanation.